Holtrop-Mennen vs CFD: Resistance of the Variant 22 Bulk Carrier

Team Aquaholics

Ashutosh Singhal (24NA10020)
Romir Vyas (24NA10058)
Ruhaan Kakar (24NA10059)

Summary

In our CFD report we noted that the results still had to be checked against an empirical method. In this report we do that with the Holtrop-Mennen method, using the same hull and the same five speeds.

At the two lower speeds, 2.3 and 4.6 m/s, Holtrop-Mennen gives about twice the CFD resistance (7.4 kN against 3.7 kN, and 31.3 kN against 14.7 kN). Nearly all of this difference comes from skin friction. The friction force in our simulation is only 12-14 % of the ITTC-1957 value. If we keep the CFD pressure force and use the ITTC friction instead, the two results are within 11 % of each other at 2.3 m/s and within 2 % at 4.6 m/s.

From 6.9 m/s upwards the results are far apart. Our drag coefficient hardly changed with speed, so the simulation did not pick up wave-making resistance. Holtrop-Mennen, on the other hand, is being used outside the range it was fitted to at these speeds. We do not think either method gives a reliable number at Fn 0.4 and 0.5.

1. Introduction

Holtrop-Mennen is a regression method based on model tests and full-scale trials carried out at MARIN in the Netherlands. It only needs the main dimensions and a few form coefficients, so it is commonly used in early design, before model tests or CFD are available. Everything it needs could be taken from our offset table and the hull data in the CFD report.

The aim of this comparison is to see where the two methods agree, where they do not, and what that tells us about our CFD setup.

2. The hull

Variant 22 is a 54 m bulk carrier with a long parallel middle body, a flat bottom and blunt ends. It has no bulbous bow. Table 1 lists the particulars used, all taken from the offset table and the CFD report.

Particular	Value	Particular	Value
Length between perpendiculars	54.0 m	Block coefficient	0.759
Breadth	11.0 m	Prismatic coefficient	0.777
Draught (even keel)	5.0 m	Midship coefficient	0.977
Displaced volume	2,254 m³	Waterplane coefficient	0.903
Wetted surface	946 m²	Centre of buoyancy	1.9 m aft of midships

Table 1. Principal particulars of Variant 22.

The prismatic coefficient of 0.777 is within the range Holtrop-Mennen was developed for (bulk carriers and tankers). Compared with those ships the hull is slightly short and shallow for its breadth (L/B = 4.9 and B/T = 2.2, where the data starts at about 5.1 and 2.4). We would therefore expect the usual accuracy of around ±10 %, not better.

3. Method

3.1 CFD

The CFD results are the ones from our earlier report: ANSYS Fluent with a two-phase air-water model, a k-ω turbulence model and 500 iterations at each speed. The hull was bare and smooth, in fresh water. Resistance was found from the converged drag coefficient and a wetted surface of 946 m² (Table 5 of that report).

3.2 Holtrop-Mennen

We worked through the eight steps of the 1982 version of the method with our own implementation, which we first checked against the worked example in the original paper. The method adds up separate parts: friction from the ITTC-1957 line multiplied by a form factor, wave-making resistance, appendages, bulbous bow, transom, and a correlation allowance for hull roughness and air resistance.

3.3 Keeping the comparison fair

To compare like with like, we used fresh water with the same density and viscosity as the CFD (998.2 kg/m³ and 1.004×10^{-6} m²/s) and the same wetted surface of 946 m². No appendages were included. The correlation allowance is listed separately, since the smooth-hull CFD has nothing that corresponds to it. The half angle of entrance (22.8°) was measured from the offsets at the design waterline, and we assumed the stern does not leave an immersed transom (see Section 7).

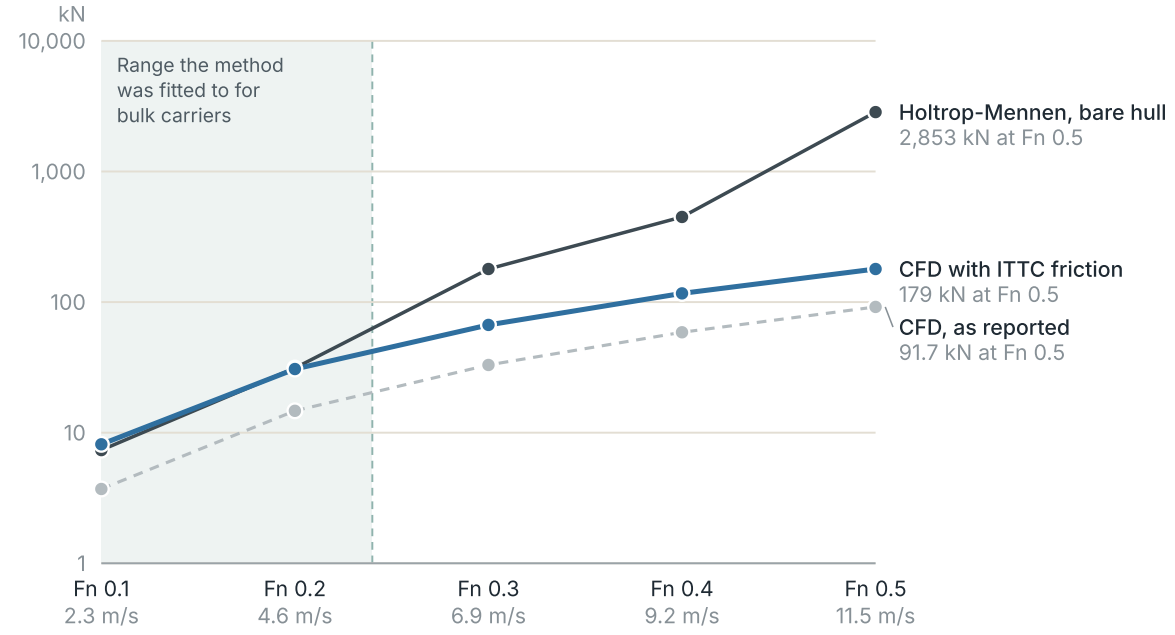
4. Results

Table 2 gives both sets of results at the five speeds. Holtrop-Mennen is higher at every speed: by about a factor of two at 2.3 and 4.6 m/s, and by much more from 6.9 m/s up.

Speed (m/s)	Speed (knots)	Froude number	CFD (kN)	Holtrop-Mennen, bare hull (kN)	Holtrop-Mennen incl. correlation allowance (kN)	Holtrop-Mennen ÷ CFD, bare hull
2.3	4.5	0.1	3.7	7.3	8.9	2.0×
4.6	8.9	0.2	14.7	31.3	37.6	2.1×
6.9	13.4	0.3	33.0	178.9	193.1	5.4×
9.2	17.9	0.4	58.7	448.7	473.9	7.6×
11.5	22.4	0.5	91.7	2,852.9	2,892.2	31.1×

Table 2. Total resistance at the five speeds. CFD values are from Table 5 of the CFD report. The ratio compares the two bare-hull values.

Figure 1. Resistance against Froude number for both methods



Logarithmic scale. "CFD with ITTC friction" is the CFD pressure force plus the ITTC-1957 friction force.

5. Discussion

5.1 Skin friction

The friction force from Fluent is 12-14 % of the ITTC-1957 value at every speed (Table 3). As mentioned in the CFD report, the mesh has no inflation layer at the hull, so the boundary layer is not resolved and the shear stress comes out too low. This alone explains most of the factor of two at the lower speeds.

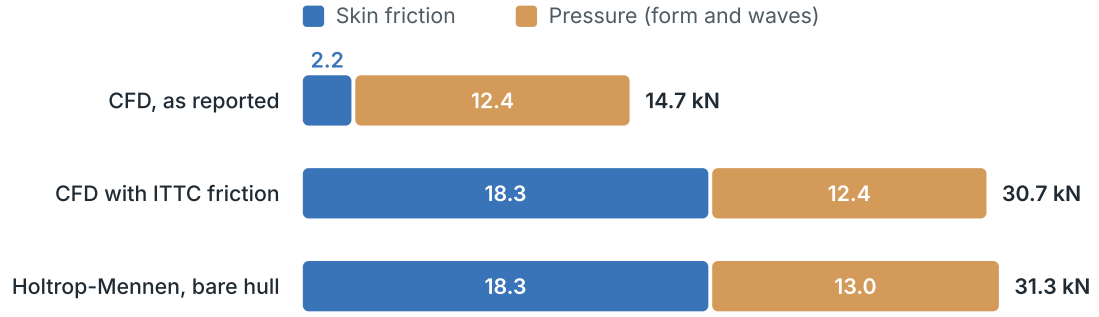
Froude number	CFD friction (kN)	ITTC-1957 friction (kN)	CFD as % of ITTC
0.1	0.6	5.05	12 %
0.2	2.2	18.33	12 %
0.3	5.1	39.06	13 %
0.4	9.0	66.88	13 %
0.5	14.0	101.54	14 %

Table 3. CFD friction force compared with the ITTC-1957 friction line for the same wetted surface.

5.2 Pressure resistance

The pressure part of the CFD result compares well with Holtrop-Mennen. At 4.6 m/s the CFD pressure force is 12.4 kN, while Holtrop-Mennen gives 13.0 kN for form and wave resistance together. At 2.3 m/s the values are 3.1 and 2.3 kN. So the high pressure share we found in the CFD seems to be mostly real, which makes sense for a full hull with blunt ends and a short run aft.

Figure 2. Friction and pressure parts at 4.6 m/s



Resistance in kN at 4.6 m/s (8.9 knots, Fn 0.2). The Holtrop-Mennen correlation allowance (6.3 kN) is not included.

5.3 Wave resistance

In the CFD the drag coefficient stays at about 1.47×10^{-3} at all speeds, so the resistance just increases with V^2. For a full hull, wave resistance should rise sharply above about Fn 0.25. At 6.9 m/s Holtrop-Mennen predicts 122 kN of wave resistance, more than four times the whole CFD pressure force. The likely reasons are the ones already listed in the CFD report: the mesh near the free surface is too coarse to capture the waves, and a steady solver was used.

5.4 Range of the method

For bulk carriers, Holtrop-Mennen was fitted only up to about Fn 0.24, and Fn 0.4 and 0.5 are well beyond the service speed of a ship like this. The 2,853 kN it gives at Fn 0.5 shows that the resistance would be very high, but it should not be read as an actual value.

6. Effective power

The effective power follows the same trend. At 4.6 m/s (about 9 knots), a realistic speed for this ship, the corrected CFD and Holtrop-Mennen both give roughly 140-175 kW, against 67 kW from the original CFD result.

Speed (m/s)	Speed (knots)	CFD as reported (kW)	CFD with ITTC friction (kW)	Holtrop-Mennen (kW)
2.3	4.5	8	19	20
4.6	8.9	67	141	173
6.9	13.4	228	462	1,332
9.2	17.9	540	1,073	4,360
11.5	22.4	1,054	2,060	33,261

Table 4. Effective power, $P_E = R_T \times V$. The Holtrop-Mennen values include the correlation allowance. Values at Fn 0.4 and 0.5 are only indicative.

7. Sensitivity of the estimate

We also checked how much the Holtrop-Mennen result changes with the assumptions we made (Table 5). Most of them change it by only a few percent. The one that really matters is the stern.

Assumption changed	R_T at 4.6 m/s (kN)	Change
None (inputs as in Sections 2 and 3)	37.6	-
Hull cut off at AP, leaving about 24 m² of immersed transom	76.2	+103 %
Wetted surface and entrance angle from the Holtrop-Mennen formulas instead of the offsets	39.9	+6 %
Sea water instead of fresh water	39.2	+4 %
U-shaped stern sections (C_{stern} = +10)	38.4	+2 %
V-shaped stern sections (C_{stern} = −10)	36.8	−2 %

Table 5. Effect of each assumption on the Holtrop-Mennen total at 4.6 m/s (Fn 0.2).

From the offsets, the hull is still 8 to 10.5 m wide at the aft perpendicular between the 3 m and 5 m waterlines. If the CAD model simply ends at AP, it has a flat transom of about 24 m² below the waterline. Holtrop-Mennen would then add around 11 kN of transom resistance at 2.3 m/s, three times our whole CFD result, and its own estimate at 4.6 m/s would roughly double. We need to check how the stern was closed in the CAD model.

On the CFD side, two different solver settings gave results 9 % apart and only one mesh was used, so the CFD values should not be quoted to more than two significant figures.

8. Conclusion

At 2.3 and 4.6 m/s, the realistic speeds for this hull, the two methods agree well once the CFD friction is corrected. The CFD captures the pressure (form) resistance reasonably well, but its skin friction is about eight times too low and it misses the wave resistance from about Fn 0.25 onwards. Our best estimate of the bare-hull resistance at 4.6 m/s is about 31 kN, or about 38 kN with the correlation allowance.

Next steps:

1. Check how the stern is closed at AP in the CAD model.
2. Add an inflation layer at the hull and rerun 2.3 and 4.6 m/s. The friction force should come out near 5 and 18 kN.
3. Run 6.9 m/s as a transient case with a finer mesh at the free surface. Holtrop-Mennen expects about 120 kN of wave resistance there.
4. Check the reference values in Fluent, as already suggested in the CFD report.
5. Treat the 9.2 and 11.5 m/s results from both methods as qualitative only.

Appendix: Holtrop-Mennen values

Table A1 lists the main terms that do not depend on speed, and Table A2 the resistance components at each speed.

Quantity	Value	Comment
Length of run, L_R	7.85 m	Short run aft
Form factor, $1 + k_1$	1.456	High, because of the short run
Half angle of entrance	22.8°	From the offsets at the design waterline
Bulb and transom factors, c_2 and c_5	1.0 and 1.0	No bulb, no immersed transom
Correlation allowance, C_A	6.30×10^{-4}	For 150 µm hull roughness

Table A1. Main Holtrop-Mennen terms for Variant 22.

Speed (m/s)	Froude number	Friction (kN)	Friction with form (kN)	Waves (kN)	Allowance (kN)	Total (kN)
2.3	0.1	5.05	7.35	0.00	1.57	8.92
4.6	0.2	18.33	26.69	4.60	6.29	37.59
6.9	0.3	39.06	56.88	122.07	14.16	193.11
9.2	0.4	66.88	97.38	351.30	25.18	473.86
11.5	0.5	101.54	147.85	2,705.05	39.34	2,892.24

Table A2. Holtrop-Mennen components at the five speeds. Appendage, bulb and transom terms are zero for this hull. Fresh water, $\rho = 998.2$ kg/m³, $\nu = 1.004 \times 10^{-6}$ m²/s.

References

1. Singhal, A., Vyas, R. and Kakar, R. (2026). *Resistance of a Bulk Carrier Hull Across Five Froude Numbers (Variant 22)*. IIT Kharagpur.
2. Holtrop, J. and Mennen, G. G. J. (1982). An approximate power prediction method. *International Shipbuilding Progress*, 29(335), 166-170.
3. ITTC (1957). Model-ship correlation line. Proceedings of the 8th International Towing Tank Conference.